exos ENT

① a) $\cos(2x) = 2\cos^2(x) - 1 = 2 \times \left(\frac{\sqrt{3}}{3}\right)^2 - 1 = 2 \times \frac{3}{9} - 1 = \frac{2}{3} - 1 = \frac{1}{3}$

b) $\cos(2x) = -\frac{7}{25}$

② a) $A(x) = \cos(6x - 7x)$ — b a — $\cos(7x)\sin(6x) - \sin 7x \cos 6x$ — b a

$= \sin(-x)$

$= -\sin x$

b) $B(x) = \cos(2x + 2x)$

$= \cos 4x$

c) $C(x) = \sin(2x + 3x)$

$= \sin 5x$

③ a) $\sin\left(x - \frac{\pi}{3}\right) = \sin x \cos\frac{\pi}{3} - \sin\frac{\pi}{3}\cos x$

$= \frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x$

$\frac{-\sqrt{3}\cos x + \sin x}{2}$

b) $\sqrt{2}\left(\cos x \times \cos\frac{\pi}{4} - \sin x \times \sin\frac{\pi}{4}\right)$

$= \sqrt{2}\left(\cos x \times \frac{\sqrt{2}}{2} - \sin x \times \frac{\sqrt{2}}{2}\right)$

$= \cos x \times \frac{\sqrt{2}^2}{2} - \sin x \times \frac{\sqrt{2}^2}{2}$

$= \cos x - \sin x$

⑥ 1) $\cos 2a = 2\cos^2\left(\frac{\sqrt{2+\sqrt{3}}}{2}\right) - 1 = \frac{\sqrt{3}}{2}$.

2) a) $0 \leq 2a \leq \pi$

donc $2a \in [0 ; \pi]$

b) $2a = -\frac{\pi}{6}$ ~~x~~ ou $2a = \frac{\pi}{6}$

$a = \frac{\pi}{12}$